Vertically Integrated Industries

Investment theses and category views on minerals, metals, manufacturing and chemistry

1. Investment theses: tailwinds, key takeaways and macro

This theme covers companies that make physical things in their own plants, such as metal parts, magnets and molecules, and that use their own software and AI to make them faster and with fewer skilled people than older competitors. They are vertically integrated because one company owns the whole job, from order to shipped product, and earns money on what it ships. The five theses below explain why demand is rising, where new companies have an edge, how buyers behave, what the unit economics look like, and where the main risk sits.

1. **The US is moving its supply of minerals, magnets and parts away from China for national security and economic resilience, and defense budgets, government lending and a lasting labor shortage will keep that demand in place for years**
   - China leads refining in 19 of 20 strategic minerals, with about 70% average share (IEA), and makes about 94% of the world's rare-earth magnets, which go into drones, missiles, motors and robots (Mining.com). The US relies fully on imports for 16 critical minerals (USGS), and because China refines most of this supply, its output and export rules set the world price. China's current pause on export controls for these materials ends on November 10 and November 27, 2026. Buyers and the government will pay for non-Chinese supply even when it costs more, because supply chain resilience is now an economic and national-security issue.
   - Defense budgets are rising and the Pentagon is paying to rebuild domestic supply of munitions and parts, with the FY27 request asking for $49.5B for 10 key munitions, about 8.5 times the average of the past decade (CNAS), and the Navy putting $900M into submarine-parts plants (Breaking Defense). Because the FY26 defense bill also lets the Pentagon commit to purchases of up to 10 years with advance payments (FY26 NDAA), factories that qualify for defense work can count on years of steady orders.
   - The government now lends to and invests in the companies rebuilding these supply chains, which lowers the cost of building new plants. The Pentagon's Office of Strategic Capital (OSC), which lends to companies in key supply chains, has made a $620M loan to Vulcan Elements, a US magnet maker, and a $500M conditional loan to Phoenix Tailings, a US rare-earth metals maker, and about $20B is requested for it in FY27 (Mintz). The Department of Energy's loan office has over $289B of lending authority (DOE), and governments worldwide put about $65B into minerals in 2025 (IEA).
   - US manufacturers will stay short of skilled workers for years, since they need to fill about 3.8M jobs by 2033 and about 1.9M of those jobs are projected to go unfilled (Deloitte), which favors plants that can run with fewer people. Of the 17,156 machine shops in the US, 83% have fewer than 20 employees (Census), and many owners are near retirement with no successor.
   - Some of the current tailwinds are temporary, so we should not build the thesis on them. Tariffs are set by proclamation and can change overnight, about 82% of the munitions request depends on a reconciliation bill Congress has not passed (CNAS), and government loans take a long time to close and come with conditions. The Fed raised rates to 3.75-4.00% in September, and spending on new factory construction was down 21.9% year on year by May 2026 (Federal Reserve, ENR). We should favor companies that still work if tariffs fall and the munitions surge slows.
2. **AI and automation let new companies quote, program and inspect parts faster and get more output from plants that run nonstop, which gives them an edge over older competitors**
   - The work around the machine includes quoting, planning, writing the instructions a machine follows (CAM programming), inspection and paperwork. Arda, a factory software startup, says its AI cut the time to write assembly instructions at Anduril from about 10 hours to 15 minutes (claim), and a former Hadrian employee told us in an expert call that Hadrian cut its lead times by about 50% through automation.
   - The machines in most plants sit idle for much of the day, which leaves room to grow output without building anything new. In a survey of more than 100 plants, cutting machines ran about 34% of the time

against a 70% target, and European process plants run at about 74% of capacity against the 80-85% they need (FTI Consulting).

- In a plant that runs nonstop, such as a chemical or metals line, every extra point of yield (good output per unit of input) is close to pure profit. Labor is under 10% of the cost of goods in a new automated plant (claim), so the case for automation rests on speed and yield more than on headcount.
- Nobody has yet shown that AI makes each unit cheaper as volume grows, so the edge today is speed, yield and the ability to grow with a smaller crew while competitors cannot hire.

3. **Defense and aerospace buyers pay a premium for fast delivery and spare capacity, and a supplier that wins approval keeps the work for decades**
   - Defense buyers care more about delivery dates than price, and a director at a defense prime told us in an expert call that "cost is secondary" because it is "driven usually by schedule", while an engineer there said he would pay 20-40% more to avoid a delay. The premium is highest on prototypes, spare parts and urgent orders, and it fades at volume.
   - A major defense contractor, often called a prime, approves a supplier only after the plant exists and is certified, which takes about nine months to two years (expert calls with two primes). Once approved, a supplier keeps the work for aerospace programs that can run about 50 years.
   - The Pentagon relies on more than 200,000 suppliers and has country-of-origin data for fewer than 10% of the lower-tier suppliers on the F-35 (GAO). Most of that supplier base is small, old and short of people, so primes need new suppliers that can deliver on time.
4. **Companies that own their plants earn the best margins, and the strongest ones get governments, customers or lenders to fund much of the build**
   - Companies that own their plants keep more of each sale than those that route work to partners, and Protolabs, a listed maker of custom parts, earns a 49% gross margin in its own plants and 31% on work it sends to partner shops (company filings). Approved incumbents also earn well, and Howmet Aerospace's engine-parts unit ran a 37.7% EBITDA margin (earnings before interest, taxes, depreciation and amortization) in the second quarter of 2026 (company filings).
   - The central risk is capital spending (capex) on buildings, machines and certifications before contracts are signed, because the money goes out years before revenue comes in. Hadrian, a precision-parts maker, committed about $200M to a single plant in Mesa, Arizona (company release), and its CEO says capex runs about two years ahead of revenue (claim).
   - The companies that succeed shift much of the cost of the build to others, using government loans and grants, purchase agreements signed before the plant is built, factories hosted inside customers' plants, bank debt once revenue exists, leased buildings, franchising, and acquisitions of certified plants.
5. **Established incumbents control supplier approvals and buy promising startups, but an aging and fragmented supplier base still leaves room for new companies**
   - The primes decide which suppliers are approved and how much work each one gets, and they cap each supplier at about a quarter to a third of their work, so even a winning supplier grows one share at a time. Precision Castparts, one of the largest aerospace parts suppliers, saw its share of Boeing's machining work fall from about 50% to 30% over ten years, and it is now automating (expert call).
   - Incumbents often buy new technology instead of building it, which gives startups buyers but also well-funded competitors. BHP's venture arm backs Mariana Minerals, Rio Tinto paid $6.7B for Arcadium Lithium, and RTX Ventures, the venture arm of the defense contractor RTX, led the seed round of Radical AI, an AI materials lab. Drug companies are also building their own AI labs, and Eli Lilly and NVIDIA plan to invest up to $1B over five years in a joint AI lab for drug discovery in the San Francisco Bay Area (Lilly and NVIDIA, January 2026).
   - Hardware companies in this space have failed when their growth ran ahead of their economics, as Markforged, a 3D-printer maker, fell from a $2.1B valuation in 2021 to a $42.5M sale in 2026 (3D Printing Journal), and Daedalus, an AI-run machining startup in Germany, went insolvent in August 2026.

- New companies win where the incumbents are weakest, among small, aging shops with no successor, new defense programs that need second sources, and plants that need software to run with fewer people.

2. Interesting categories

The categories below follow our primer, and for each one we set out the main tailwinds and headwinds and state whether we should focus there, judged on the category as a whole.

Fabrication and Precision & Defense are both full-stack factories that own the plant, the software and the operating model, so the difference between them is the buyer and the qualification bar. If a part needs aerospace or defense certification such as AS9100 or ITAR, it belongs in Precision & Defense, and if it is an industrial part bought mainly on speed and cost, it belongs in Fabrication.

A. Raw Materials

Stage A: Locating & finding minerals and mines

- **Tailwinds: Cutting reliance on China starts with finding new deposits of copper, lithium, nickel and rare earths in the US and allied countries.** The US relies fully on imports for 16 mineral commodities and for more than half of 54 others, including 67% of its rare earths and 57% of its refined copper (USGS).
- **Demand for these metals is rising with batteries, magnets, the grid and data centers.** Copper demand alone is expected to rise about 50% by 2040 as these uses grow (Robeco).
- **AI combined with cheaper drilling and faster lab testing lowers the cost of checking each exploration target.**
- **Headwinds: Most exploration projects never become producing mines, and the few that do take 15 to 19 years.** Only about 0.5% of exploration projects succeed by one estimate (claim), and mines average 15.7 years from discovery to production, US mines opened in 2020-24 took 19.1 years, and US permitting alone commonly takes 7-10 years, against about 2 in Australia and Canada.
- **Each discovery is expensive for the industry as a whole, and few of the deposits found become mines.** The industry spends about $300M on exploration per deposit found, and only about a fifth of the roughly 40 deposits found each year become mines (Earth AI, claim).
- **Why we should not focus here: Explorers rarely keep the value of what they find, because the miner that builds the mine takes most of the economics.** Explorers usually bring in a miner through an option or joint venture, in which the miner funds the drilling to earn 51-70% of the project. The explorer keeps a minority stake or a royalty, usually a net smelter return (NSR) of 1-3% of metal sales after transport (industry option terms). An explorer that keeps the mine must raise billions, since a new underground copper mine can cost over $2B (Ecofin), and then wait 15 to 19 years for production. In both cases the return is too small, arrives too late or needs too much capital to justify an early equity investment.

Example companies: Earth AI, KoBold Metals, Terra AI, GeologicAI, Ideon

Stage B: Mining & refining

- **Tailwinds: China controls most of the world's refining and magnet making, so buyers and governments are paying to build that capacity outside China.**
- **China also uses export controls to cut supply to other countries, which drives prices up outside China and shows buyers what depending on it costs.** China has required export licenses for gallium and germanium since August 2023, graphite since December 2023, antimony since September 2024 and seven rare earths since April 2025, and in December 2024 it banned gallium, germanium and antimony sales to the US outright, a ban it has paused until November 27, 2026 (WTO, MOFCOM). Antimony in Rotterdam rose from about $22,500 a tonne in July 2024 to about $58,000 in April 2025 (Fastmarkets), and germanium metal in Europe rose from $3,150 to $5,380 per kilogram between January and October 2025 (USGS). Planned non-

Chinese magnet capacity covers only about a third of expected 2035 demand (IEA), so buyers will keep paying for new supply.

- **Government money lowers the cost of building new refineries and magnet plants in the US.** Pentagon loans and 10-year purchase commitments with price floors support new plants, and the 45X tax credit covers about 10% of production cost until it phases out in 2031-34 (CRS).
- **The US is rebuilding its own uranium enrichment because a 2024 law bans Russian enriched uranium and new reactors need a fuel that only Russia has sold at commercial scale.** The Prohibiting Russian Uranium Imports Act bans imports of Russian low-enriched uranium (LEU), the fuel for today's reactors, and all waivers end by January 1, 2028 (Public Law 118-62, DOE). Most advanced reactor designs need HALEU (high-assay low-enriched uranium, enriched to between 5% and 20%), which no US supplier yet sells and which has been commercially available only from Russia (DOE, Reuters). In January 2026 the Department of Energy awarded $2.7B of contracts to rebuild domestic enrichment, including $900M to General Matter, a startup founded in 2024 that is building a plant in Paducah, Kentucky and says it will enrich uranium by the end of the decade (DOE, claim).
- **Headwinds: Western plants take longer and cost more to build than Chinese ones, since Chinese refineries take about 6 months to commission, against 2 to 4 years in the West.** A Western plant takes about 5 years to build and another 3 to 5 to ramp after permits (claim).
- **China sets the world price of these materials because it refines most of the supply, so the output and costs of Chinese plants decide what every other producer is paid.** China refines over 95% of battery-grade graphite and magnet rare earths, over 75% of cobalt and about 70% of lithium chemicals (IEA), and it makes 99% of the world's primary gallium (USGS). The market price for the main magnet rare earths is set in China (Reuters), and the same holds across these materials, so a US plant sells at the Chinese price whatever its own costs are.
- **Chinese producers, many of them backed by the state, have kept adding supply as prices fell, which has pushed prices below what a new Western plant needs to break even.** Lithium carbonate in China fell from about 597,500 yuan a tonne in November 2022 to under 78,000 in January 2025, a drop of about 86% (Policy Center for the New South, Reuters), and Albemarle paused its $1.3B South Carolina lithium refinery in part because of overproduction by Chinese rivals (Reuters). The Chinese price of NdPr, the main magnet rare earth, fell from about $60 to $44 per kilogram between June 2023 and June 2024 (Lynas), and MP Materials, the largest US rare-earth producer, lost $85.9M on $224.4M of revenue in 2025 (company filings). The Commerce Department found Chinese battery graphite subsidized at about 67% and sold in the US at dumping margins of 93.5% (Commerce).
- **Why we should focus here: Mining and refining have the most lasting demand in the theme, because the need for non-Chinese metal and magnets comes from national security and supply chain resilience.** New companies have a real edge here, because restarting old mines that already have permits shortens the timeline, processing scrap and mine waste avoids a new mine altogether, and AI-run process control raises yield in plants that run around the clock. Because a US plant cannot win on Chinese prices alone, the companies to back secure a price floor, long contracts with buyers that pay extra for non-Chinese supply, tariff protection or a government buyer before they build. The Pentagon set that model in July 2025, when it guaranteed MP Materials $110 per kilogram of NdPr for 10 years, nearly twice the Chinese market price at the time (MP Materials, Reuters). Protected revenue of this kind makes the category worth backing now despite the pricing risk.

Example companies: Mariana Minerals, BHP, Rio Tinto, Caterpillar, Vulcan Elements, Phoenix Tailings, ReElement, Ucore, Energy Fuels, MP Materials, General Matter, Hertha, Cyclic, Mass Magnetics, Duranium, Boston Metal, Electra, Ceibo, Nucor, Carpenter Technology

B. Manufacturing

Metal Supply

- **Tailwinds: Tariffs, buy-American rules and reshoring are raising demand for US metal from machine shops, fabricators and primes.** Since April 2026 the Section 232 tariffs have charged 50% on imported metal articles and 25% on goods that contain steel, aluminum or copper (White & Case), which pushes buyers toward domestic stock.
- **Headwinds: Established distributors already serve this market well and are gaining share without AI, which leaves little room for a new entrant.** The largest is Reliance, Inc., with $14.3B of net sales in 2025, about 310 locations and about 17% of all tons sold by US service centers (Reliance 10-K). Reliance grew its tons sold by 6.2% in 2025 while the industry shrank by 1.0%.
- **Distribution earns thin margins even at the largest scale, which leaves a newcomer little room to pay for technology.** Distributors earn a spread on the metal plus cutting fees and hold much of their cash in inventory, so even at Reliance's scale the pretax margin was 6.8% in 2025, and Ryerson, the second largest, posted a net loss.
- **Why we should not focus here: The margin in metal supply stays with the incumbents, who already have the stock, the buying power and the customers.** A new distributor ties up its capital in inventory and competes on price against bigger incumbents. The main tailwind is a tariff, and a margin that rests on a tariff can disappear with one policy change.

Example companies: Nox Metals, Reliance, Ryerson

Fabrication

- **Tailwinds: Full-stack fabrication companies own the plant, write the software that runs it and set the operating model, so they make general industrial parts fast and in volume and keep the whole margin on each job.** The model case is 1872, a Cincinnati steel fabricator whose software turns a customer's design file into a build plan and schedule for its robots, and which says it builds "the full stack," meaning the factory, the software and the operating model as one system (1872). The parts are sheet metal, welded steel frames and skids, circuit boards, control panels and molded parts, sold to energy, utility, industrial and consumer buyers who choose on speed and price. None of this work needs aerospace or defense certification, and that is the test separating Fabrication from Precision & Defense.
- **Reshoring is bringing new assembly plants to the US, and each one needs local suppliers of frames, panels and boards.** Companies announced about 247,000 US jobs from reshoring and foreign investment in 2025 (Reshoring Initiative).
- **The shortage of welders and machinists favors plants that automate cutting and bending instead of hiring more people.**
- **Headwinds: The speed premium fades at volume, and US shops beat Chinese prices on only part of their work (claim).** A former Anduril engineer told us that parts from Taiwan cost only about 20-30% less than US parts and arrive more reliably (claim).
- **Reshoring announcements run well ahead of real orders, so fabricators cannot plan capacity on headlines.** Only 14 of 64 conditional reshoring announcements had turned into actual projects by mid-2026 (Reshoring Initiative).
- **Customers that grow often buy their own machines and take the work in-house.**
- **Why we should focus here: Full-stack fabrication plants cost far less than precision plants and keep the whole margin on each part, so companies in this category can grow on their own revenue and reach profit early.** Owning the factory, the software and the customer lets a company price and schedule every job itself, and Protolabs, the closest listed comparison, keeps more margin in its own plants than on work it sends to partner shops. Buyers that pay for speed, low capital needs and early profits make this one of the most attractive categories in the theme.

Example companies: 1872, SendCutSend, CircuitHub, Podium Automation, Atomic Industries

Precision & Defense Hardware

- **Tailwinds: Precision & Defense factories make flight-critical and defense hardware that needs aerospace or defense certification, and the Pentagon needs a larger base of trusted domestic suppliers for it.** These are metal or composite parts made to tight tolerances by machining, 3D printing, casting or molding for missiles, jets, satellites, submarines and chip equipment, and each one needs AS9100 or ITAR approval plus a qualification at the buyer that takes months. The buyers are aerospace, defense, space and semiconductor companies, on the commercial side as well as the defense side. Munitions and submarine budgets are rising, with purchase commitments of up to 10 years now allowed.
- **Commercial aerospace adds a second and steadier source of demand, because Airbus and Boeing together hold orders for nearly 15,000 aircraft and parts supply sets how fast they can build.** Airbus ended 2025 with a record backlog of 8,754 aircraft and Boeing with more than 6,100 (company results). A Pratt & Whitney engine shortfall delays Airbus's main jet program from reaching 70 to 75 a month until the end of 2027 (Airbus FY2025 results). Suppliers are already growing on this demand, and Howmet's revenue rose 24% in the second quarter of 2026, with commercial aerospace up 28%, defense aerospace up 11% and gas turbines up 38% (Howmet results).
- **Power, chip and space hardware need the same precision parts and are growing as fast as aerospace.** GE Vernova has sold out its gas turbines through 2028 and had 122 GW under contract in the third quarter of 2026 (GE Vernova), chip equipment sales are forecast to reach a record $165.9B in 2026, up 23% (SEMI), and the world made a record 329 orbital launch attempts in 2025 (planet4589). Medical devices, EVs and robot joints need the same tight-tolerance metal parts, which widens the buyer base further.
- **Headwinds: A new precision plant has to be built and certified before it wins volume work, so capex runs well ahead of revenue.** AS9100, the aerospace quality standard, typically takes 6-12 months, and approval at each buyer takes about nine months to two years.
- **Commercial work brings its own approvals, because aircraft and engine makers also require Nadcap accreditation for special processes.** Nadcap is an industry-run program that audits processes such as heat treating, welding and inspection.
- **Commercial buyers push hard on price, and a few buyers control most of the volume.** Spirit AeroSystems, Boeing's largest structures supplier, booked $585M of losses on fixed-price contracts in one quarter of 2025 (Reuters), and Boeing bought it back that December (Boeing). Two airframers and a few engine makers place most orders, so a supplier tied to one carries its problems.
- **Defense money arrives slowly and depends on Congress, even in years when budgets rise.** Carpenter Technology reported that its defense bookings fell because customers were "not receiving funds" (Carpenter earnings call).
- **Some companies sell their robots as well as their parts, which blurs whether the value sits in the factory or in the equipment.** Machina Labs is a hybrid that makes metal structures in its own plants and also places its robot cells at customer sites (Machina Labs).
- **Why we should focus here: An approved supplier keeps the work for programs that can run 50 years, primes pay a premium for fast delivery, and approved incumbents earn strong margins, so the prize at the end of approval is real.** New companies can reach it by building their own plants, by buying certified shops from retiring owners at about seven times annual earnings (claim), or by hosting cells inside a buyer's plant, and each model trades some margin for less capital.
- **The strongest new precision companies sell to both defense and commercial buyers, because commercial aerospace, power and chip equipment bring larger and steadier volumes that do not depend on appropriations.** That mix keeps machines busy between defense orders and spreads the cost of approvals.

Example companies: Hadrian, Isembard, Divergent, Freeform, Machina Labs, Senra Systems, Foundry Robotics, Amca, Fabri, Layup, Hybron, Rangeview, Furientis, Helicon, Westmag, Re:Build, Anduril, Precision Castparts, Howmet

C. Specialty Chemicals

Producers (hybrid)

- **Tailwinds: Drug companies are moving discovery chemistry out of China because relying on it has become a supply chain, intellectual property and national-security concern.** BIOSECURE, a law passed in December 2025, limits the use of certain Chinese biotech suppliers (Baker McKenzie), and 79% of 124 biotechs surveyed had a China-based contract manufacturer (BIO).
- **The government is adding pressure on the largest Chinese chemistry suppliers that US drug companies use.** In June 2026 the Pentagon added WuXi AppTec, the largest Chinese drug research and manufacturing firm, to its list of Chinese military companies (Reuters).
- **AI that designs molecules and robots that make them bring down the cost of doing this work in the US and Europe.** Chemify, a Glasgow-based lab, says it already works with 8 of the 20 largest drug companies (claim, SCI).
- **Headwinds: US chemists cost several times Chinese rates, so these labs have to automate to reach offshore prices.**
- **The category is young, labs have not disclosed revenue, and drug companies are building their own AI labs.**
- **The legal push away from China can soften, which would weaken the main reason to switch suppliers.** In August 2026 a federal judge blocked the Pentagon's listing of WuXi AppTec while the case proceeds (Reuters).
- **Why we should focus here: Drug companies want a domestic or allied source for discovery chemistry, and they now have a legal reason to move work out of China.** The automated labs in this category make new molecules at milligram-to-gram scale and do not develop their own drugs. They are paid per compound delivered, or through milestones and royalties if a molecule advances. If automation brings unit cost close to offshore prices, the category can take a meaningful share of work that is leaving China.

Example companies: Chemify, Warp Sciences

D. Advanced Materials

Materials Discovery Labs

- **Tailwinds: AI models and robotic labs can compress the long and costly path from a new material idea to a working product, which is the slowest step in the whole materials chain.** A new material has typically taken 10 to 20 years and more than $100M to develop (claim), and the leading labs report making more than a thousand alloys in about six months (claim). Simulation is getting cheaper as well, and a national-lab machine-learning framework runs electronic-structure calculations more than 5,000 times faster than the standard method on 2,000-atom systems (DOE). Open datasets of more than 100 million materials calculations are now free for any lab to train its models on (industry research).
- **Chips, data centers, batteries, magnets and defense systems all need materials that do not exist today, so a lab with a working discovery already has buyers.** Data centers are on track to use about 945 TWh of power by 2030, up from 415 TWh in 2024, and cooling takes 7% to over 30% of a site's power (IEA). Chipmakers need new ways to carry heat away from denser chips, and AI materials labs already report paid work for semiconductor makers on heat dissipation (claim).
- **Battery, magnet and defense buyers now treat materials free of Chinese inputs as a security goal, which gives discovery labs a clear target in substitutes built from elements available elsewhere.** China makes about 85% of battery cathode material and over 98% of the cheaper LFP type, and the IEA expects China still to hold about three-quarters of cathode production in 2035 (IEA).
- **The federal government is putting the national laboratories behind AI-driven materials work, which gives labs public partners, computing and federal datasets to draw on.** The Genesis Mission, launched by executive order in November 2025, joins DOE's 17 national laboratories in one AI platform for science, and

one of its 26 challenges is to design materials from performance goals and shrink development timelines from decades to months (DOE). DOE also lists faster development of defense-ready materials among the mission's national security goals (DOE).

- **Major investors and industrial companies are funding discovery labs early, which pays for the robotic labs and data these companies need long before they earn revenue.** Leading AI materials labs raised seed and early rounds of $55M to $550M in 2025, with chipmakers and a defense prime among the backers (company announcements).
- **Headwinds: A discovery is only the start, because scaling a new material up and getting customers to qualify it still takes years after the lab finds it.** Customers take one to five years to qualify a new material, and finding a material faster does not shorten that qualification time.
- **Most labs reach volume through a partner that scales the material up and sells it, which leaves the lab a fee or royalty and little of the margin on each ton.** These labs do not yet sell material at volume, and their revenue today comes from paid R&D, licensing and research awards of around $1M, which is small next to their valuations.
- **Few AI-predicted materials have been made and proven useful, so the published counts of new materials overstate what has reached the lab bench.** Academic chemists who reviewed one widely cited set of 2.2 million AI-predicted crystals found scant evidence of compounds that were novel, credible and useful at once (peer-reviewed critique). A second review argued that none of the materials first reported by a leading autonomous lab were new, and the original paper was later corrected from 41 compounds out of 58 targets to 36 out of 57 (peer-reviewed critique, journal correction).
- **Big technology companies run their own AI materials programs and publish models and data openly, which gives buyers and incumbents a free starting point for in-house work.** Big technology companies have published millions of predicted crystals and released open generative design models, and one such model designed a compound that was made in the lab and came within 20% of its target stiffness (industry research).
- **Discovery labs need hundreds of millions of dollars and many years before revenue, so returns depend on a few discoveries reaching volume.** The best-funded labs have raised $300M to $550M each while revenue stays small (company announcements), and a lab that builds its own plant to keep more of the margin takes on plant costs of hundreds of millions of dollars, where even a leading European cathode maker's battery-materials unit lost money on €436M of revenue (company filings).
- **Why we should focus here, on a longer time horizon: If these labs work, they shorten the slowest part of the materials chain, which is finding a material worth qualifying.** Orbital Materials, for example, pairs generative AI with its own lab experiments to design new materials for commercial use, and has found new molecular classes for cooling high-density GPUs in data centers (Orbital).
- **The business model decides how much of that value a lab keeps, so we should favor labs that pair paid R&D and licensing today with a path to selling material by the kilogram or ton under supply deals.** Contract manufacturing keeps the capital need of that path far below that of a refinery, and a first material with a paying industrial customer, such as a chipmaker or a data-center operator, shortens the road to those deals. Today the category is more of an experiment for us, with a longer time horizon than the factory categories above.

Example companies: Orbital Materials, Radical AI, CuspAI, Lila Sciences, Periodic Labs, Aionics, Mattiq, Dunia Innovations, Entalpic, Discovered Materials, Chemix

E. Enabling Software & Robotics

Enablers

- **Tailwinds: Reshoring only works if US plants can run with fewer skilled people, because about 1.9M US manufacturing jobs are projected to go unfilled by 2033 (Deloitte).** Fully loaded US factory labor costs

about $46.30 an hour, while the price of an industrial robot fell from about $46,000 in 2010 to about $23,000 in 2022 (Emergence Capital), so automation pays back faster each year.

- **China installs more industrial robots than the rest of the world combined, and its lead is growing.** China installed 354,226 industrial robots in 2025, 59% of the 603,307 installed worldwide and up 20% from a year earlier, while the US installed 38,428, up 12% (IFR).
- **China is also building its own robot industry, so it is less dependent on Western suppliers each year.** Chinese brands supplied about 30% of robots installed in China in 2019-21 and more than half since 2024 (IFR), and a 2023 government plan for humanoid robots set targets for mass production by 2025 and a secure domestic supply chain by 2027 (MIIT).
- **Washington is starting to shut Chinese robots out of sensitive US sites, which gives US robot and software makers a protected market.** In June 2026 the Pentagon added the Chinese robot maker Unitree to its list of Chinese military companies (Reuters), and on July 28, 2026 the FCC added foreign-made mobile and humanoid robots, including those from allied countries, to its Covered List, which blocks new models from US sale without a conditional approval (Cleary Gottlieb). The House-passed FY27 defense bill would also bar the Pentagon from buying humanoid robots made or controlled by China (H.R. 8800, Sec. 163), though that provision is not yet law.
- **Headwinds: Sales to plants are slow because manufacturers spend cautiously and take months to approve new systems.** Manufacturers expect to raise capital spending by only 2.6% over the next year, and 63% plan to import machinery (NAM).
- **Incumbent vendors such as Siemens and Rockwell Automation bundle software with hardware they have already installed.** Rockwell earns a 49.5% gross margin (Rockwell results), so a startup has to beat a vendor the plant already trusts.
- **Robots are not yet reliable enough for most factory work, since robot-learning success is 59-66% on short tasks against the 99.9% or more that a factory needs, and most humanoids remain pilots (IFR).**
- **The robot arm is only part of the cost, because integrators put the arm at roughly a fifth to under half of an installed cell, with fixtures, safety and programming making up the rest (integrator cost breakdowns).**
- **US startups depend on foreign robot hardware, which limits how much of each sale they keep.** Most robots used in the US are imported from Japan and Europe (IFR), and the German maker KUKA has been owned by China's Midea since 2017 (Reuters).
- **Why we should focus here: China's robot lead means its factories get cheaper and faster every year, so the cost gap with US plants widens unless US plants automate too.** BCG finds automation can cut labor needs by 30-60%, yet a new aerospace and defense plant still costs less to run in China (BCG).
- **Reshoring only pays with automation, because the US lacks the workers and the low wages to copy China's labor-heavy factories.** Much of the returning work is high-mix and low-volume, with defense shops running batches of 10 or 20 parts (NDIA), so the winning systems make a small batch worth automating.
- **The US opportunity is in the software, AI and integration that sit on top of the robot, because the arms themselves come from Japan, Europe and China.** The products that matter program robots from a drawing, schedule plants, inspect parts and run robot cells as a service. Security rules add demand for this layer, because defense plants need software and connected robots that no Chinese company can reach remotely.
- **Buyers pay when the system adds output from machines they already own or covers a shift they cannot staff.** By our estimate, one operator on two shifts costs about $185,000 a year, so a $350,000 cell covering both shifts pays back in about two years.
- **Software and robotics companies grow faster than plant owners because they do not build factories, and we should back those paid on measured output, starting in defense and aerospace shops.**

Example companies: Lumaril, Arda, Mind Robotics, GrayMatter Robotics, CloudNC, AI Robotics, Bright Machines, Rapta, Allus, Faraday, Project Prometheus, Unlimited Industries, Siemens, Rockwell Automation, Honeywell, AspenTech, Emerson